US Radio Frequency Cheat Sheet

What each range is called, what it's used for, and what license you need to transmit on it.

The radio spectrum at a glance

LF	MF	HF	VHF	UHF	SHF	EHF
30–300 kHz	**300 kHz–3 MHz**	**3–30 MHz**	**30–300 MHz**	**300 MHz–3 GHz**	**3–30 GHz**	**30–300 GHz**
Low	Medium	High	Very high	Ultra high	Super high	Extremely high
Navigation, 2200 m ham	AM radio, 160 m ham	Shortwave, CB, long-range ham	FM radio, air, marine, 2 m ham	TV, GMRS, cell, Wi-Fi, 70 cm ham	5 and 6 GHz Wi-Fi, satellite	Radar, short 5G links

Convert frequency to wavelength: meters = 300 ÷ MHz. So 146 MHz is about 2 m, and 446 MHz is about 0.67 m, which is rounded to "70 cm." Higher frequencies usually travel shorter distances and have shorter antennas. HF can bounce off the ionosphere, while VHF and up are mostly line of sight.

License types

Group	License type	What it means	Examples
Anyone	**No license**	Low-power devices that are FCC certified.	Wi-Fi, Bluetooth, LoRa
Anyone	**Licensed by rule**	No application or fee, but the service rules still apply.	FRS, MURS, CB, US boats and aircraft
Anyone	**Listen only**	Nothing to transmit on. Receiving is free.	NOAA Weather, GPS, WWV
Personal	**GMRS**	Application and fee, no test.	GMRS radios and repeaters
Personal	**Ham**	Test required. Class sets your frequencies.	Technician, General, Extra
Organization	**FCC station or carrier**	Issued to an agency or company for set frequencies.	Police, radio stations, cell carriers
Government	**Federal only**	Assigned by NTIA, not the FCC. Not public.	Military, FBI, FAA systems

Services you can actually use

Service	Frequencies	Channels	Max power	License and cost	Key rules
FRS	462 and 467 MHz	22	2 W ERP; 0.5 W on ch 8–14	None	No repeaters. Attached antenna only.
GMRS	462 and 467 MHz	30	50 W, 5 W or 0.5 W by channel	$35, no test. 10 years. Age 18+.	Repeaters allowed. Covers immediate family. Give call sign every 15 minutes and at the end.
MURS	151.82–154.60 MHz	5	2 W	None	No repeaters. No call sign.
CB	26.965–27.405 MHz	40	4 W AM; 12 W PEP SSB	None	No call sign required.
Ham	135.7 kHz and up	Open	1,500 W PEP; lower limits on some bands	Test, $35. 10 years.	Give call sign every 10 minutes and at the end. No business use and no encryption.
Marine VHF	156–162 MHz	Set list	25 W on ships	None for US boats	For use on vessels. Monitor ch 16 when underway.

Power figures and license rules come from 47 CFR Parts 80, 95 and 97. Use a radio certified for the service: ham radios generally aren't legal on FRS, GMRS, MURS or CB.

Rules worth remembering

- **Listening:** no license is needed to receive, but federal law restricts intercepting cell calls and some private or encrypted traffic, and some states limit scanners in vehicles.
- **Transmitting:** a radio being able to tune to a frequency doesn't make it legal to use. You need the right license and equipment for that service.
- **Emergencies:** 47 CFR 97.403 lets a ham station use any means of radio communication for immediate safety of life when normal communication isn't available. It doesn't cover practice or convenience.

Master frequency list, low to high

Common US uses. Many ranges overlap or are shared. Slashes separate paired ranges, one for each direction.

Frequency	Called	Used for	License to transmit	FCC Part
LF and MF: 30 kHz to 3 MHz				
135.7–137.8 kHz	**2200 m ham**	Experimental, long range at night	Ham: General+	97
472–479 kHz	**630 m ham**	CW and data, 5 W EIRP limit	Ham: General+	97
535–1705 kHz	**AM broadcast**	Radio stations, 540–1700 kHz	FCC broadcast license	73
1.800–2.000 MHz	**160 m ham**	"Top band," regional at night	Ham: General+	97
HF: 3 to 30 MHz (shortwave)				
3–30 MHz, scattered	**Shortwave**	Broadcast, marine, aviation, military	Varies by user	Various
2.5, 5, 10, 15, 20 MHz	**WWV**	NIST time and frequency signals	Listen only	Federal
3.500–4.000 MHz	**80 m and 75 m ham**	Regional, best at night	Tech (part), General+	97
5.3515–5.3665 MHz + 4 ch	**60 m ham**	Shared secondary band, regional	Ham: General+	97
7.000–7.300 MHz	**40 m ham**	Regional by day, long range at night	Tech (part), General+	97
10.100–10.150 MHz	**30 m ham**	CW and data only	Ham: General+	97
14.000–14.350 MHz	**20 m ham**	Main long-range daytime band	Ham: General+	97
18.068–18.168 MHz	**17 m ham**	Long range	Ham: General+	97
21.000–21.450 MHz	**15 m ham**	Long range, follows the solar cycle	Tech (part), General+	97
24.890–24.990 MHz	**12 m ham**	Long range, follows the solar cycle	Ham: General+	97
26.965–27.405 MHz	**CB (11 m)**	40 channels, truckers and local chat	Licensed by rule	95 D
28.000–29.700 MHz	**10 m ham**	Technician voice on 28.3–28.5 MHz	Tech (part), General+	97
VHF: 30 to 300 MHz				
30–50 MHz	**VHF low band**	Land mobile: public safety, utilities	FCC station license	90
50–54 MHz	**6 m ham**	"Magic band," sudden long-range paths	Ham: Technician+	97
54–88 MHz	**TV channels 2–6**	VHF-low television	FCC broadcast license	73
72–73, 75.4–76 MHz	**Radio control**	Models. 75 MHz is surface models only	Licensed by rule	95 C
88–108 MHz	**FM broadcast**	Radio stations, noncommercial below 92	FCC broadcast license	73
108–118 MHz	**Air navigation**	VOR and ILS localizer signals	Listen only	87
118–137 MHz	**VHF airband**	Pilots and towers, AM. Ground needs license	By rule (US aircraft)	87
137–138 MHz	**Satellite downlinks**	Weather and messaging satellites	Listen only	Various
138–144 MHz	**Federal land mobile**	Military and federal agencies	Federal only	NTIA
144–148 MHz	**2 m ham**	Most popular ham band, many repeaters	Ham: Technician+	97
148–150.8 MHz	**Federal land mobile**	Military and federal agencies	Federal only	NTIA
150.8–174 MHz	**VHF high band**	Police, fire, business. 162–174 mostly federal	FCC station license	90
151.82–154.60 MHz	**MURS**	5 business and personal channels, 2 W	Licensed by rule	95 J
156–162 MHz	**Marine VHF**	Boats, Coast Guard. Shore needs license	By rule (US boats)	80
160.2–161.6 MHz	**Railroad**	AAR channels: dispatch and crews	FCC station license	90
162.400–162.550 MHz	**NOAA Weather Radio**	7 channels of forecasts and alerts	Listen only	Federal
174–216 MHz	**TV channels 7–13**	VHF-high television	FCC broadcast license	73
219–220, 222–225 MHz	**1.25 m ham**	219–220 is digital forwarding only	Ham: Technician+	97
225–300 MHz	**Military airband**	Continues up to 400 MHz, uses AM	Federal only	NTIA
UHF: 300 MHz to 3 GHz				
300–400 MHz	**Military airband**	Military aircraft and federal systems	Federal only	NTIA
406.0–406.1 MHz	**Distress beacons**	EPIRB, PLB and ELT, emergency use only	Registered beacon	80, 87, 95
406.1–420 MHz	**Federal land mobile**	FBI, military and other federal agencies	Federal only	NTIA
420–450 MHz	**70 cm ham**	Handhelds, repeaters, satellites	Ham: Technician+	97
450–470 MHz	**UHF land mobile**	Police, fire, business (not GMRS/FRS)	FCC station license	90
462.5500–467.7125 MHz	**FRS**	22 channels shared with GMRS, up to 2 W	Licensed by rule	95 B
462.5500–467.7250 MHz	**GMRS**	30 channels, up to 50 W, repeaters allowed	GMRS (no test)	95 E
470–512 MHz	**T-band**	Land mobile in certain large cities	FCC station license	90
470–608 MHz	**TV channels 14–36**	UHF TV. Ch 37 is radio astronomy	FCC broadcast license	73
617–652 / 663–698 MHz	**600 MHz cellular**	5G and LTE low band	FCC carrier license	27

Frequency	Called	Used for	License to transmit	FCC Part
698–758 / 776–787 MHz	**700 MHz cellular**	LTE low band	FCC carrier license	27
758–768 / 788–798 MHz	**FirstNet broadband**	Public safety cellular network	Public safety licensee	27, 90
769–775 / 799–805 MHz	**700 MHz public safety**	Police, fire and EMS radio systems	Public safety licensee	90
806–824 / 851–869 MHz	**800 MHz land mobile**	Trunked public safety and business	FCC station license	90
824–849 / 869–894 MHz	**850 MHz cellular**	Original US cellular band	FCC carrier license	22
896–901 / 935–940 MHz	**900 MHz land mobile**	Business and industrial	FCC station license	90
902–928 MHz	**33 cm ham**	Shared with unlicensed devices	Ham: Technician+	97
902–928 MHz	**900 MHz ISM**	LoRa, Meshtastic, smart meters	No license	15
960–1215 MHz	**Aviation systems**	Transponders, ADS-B on 978 and 1090 MHz	Aviation systems only	87
1176–1227, 1575.42 MHz	**GPS**	L5 1176.45, L2 1227.60, L1 1575.42 MHz	Listen only	Federal
1240–1300 MHz	**23 cm ham**	Digital voice, video, satellites	Ham: Technician+	97
1710–1780 / 2110–2200 MHz	**AWS cellular**	LTE and 5G mid band	FCC carrier license	27
1850–1990 MHz	**PCS cellular**	LTE and 5G mid band	FCC carrier license	24
2300–2310, 2390–2450 MHz	**13 cm ham**	Data, video, satellites	Ham: Technician+	97
2400–2483.5 MHz	**2.4 GHz ISM**	Wi-Fi, Bluetooth, microwave ovens	No license	15
2496–2690 MHz	**BRS and EBS**	5G mid band	FCC carrier license	27
SHF and EHF: 3 GHz and up				
3300–3450 MHz	**9 cm ham**	Being phased out. Check current rules	Ham: Technician+	97
3450–3550 MHz	**3.45 GHz cellular**	5G mid band	FCC carrier license	27
3550–3700 MHz	**CBRS**	Citizens Broadband, shared 5G	Registered devices	96
3700–3980 MHz	**C-band cellular**	5G mid band	FCC carrier license	27
5150–5895 MHz	**5 GHz Wi-Fi**	Wi-Fi 5 and 6	No license	15
5650–5925 MHz	**5 cm ham**	Data links and microwave experiments	Ham: Technician+	97
5925–7125 MHz	**6 GHz Wi-Fi**	Wi-Fi 6E and 7	No license	15
10.0–10.5 GHz	**3 cm ham**	Microwave experiments	Ham: Technician+	97
10.7–12.7 GHz	**Ku-band satellite**	Satellite TV and internet downlinks	FCC carrier license	25
24.00–24.25 GHz	**1.2 cm ham**	Microwave experiments	Ham: Technician+	97
47 GHz and up	**Millimeter ham bands**	47.0–47.2, 76–81, 122.25–123, 134–141, 241–250, and 275+ GHz	Ham: Technician+	97

Key frequencies to memorize

Frequency	What it is
121.5 MHz	Civil aviation emergency
243.0 MHz	Military aviation emergency
156.800 MHz	Marine ch 16: distress and calling
156.525 MHz	Marine ch 70: digital distress calling
156.650 MHz	Marine ch 13: bridge-to-bridge
157.100 MHz	Marine ch 22A: Coast Guard liaison
161.975, 162.025 MHz	AIS ship tracking
27.065 MHz	CB ch 9: emergencies
27.185 MHz	CB ch 19: truckers, highway
462.675 MHz	GMRS ch 20: informal emergency use
406 MHz	EPIRB, PLB and ELT distress beacons

Ham FM and repeaters

Band	Simplex calling	Usual repeater offset
6 m	52.525 MHz	−1 MHz
2 m	146.520 MHz	± 600 kHz
1.25 m	223.500 MHz	−1.6 MHz
70 cm	446.000 MHz	± 5 MHz
GMRS	None official	+5 MHz (462 out, 467 in)

NOAA Weather Radio (MHz)

162.400, 162.425, 162.450, 162.475, 162.500, 162.525, 162.550

Ham license classes

Class	Exam	Questions	To pass	What it opens
Technician	Element 2	35	26	All ham bands from 50 MHz up, plus small HF segments
General	Element 3	35	26	Most of HF, plus 2200 m and 630 m
Amateur Extra	Element 4	50	37	Every amateur frequency

Tests are taken in order. There's no Morse code test. New licenses cost $35 and renewals $35, while upgrades have no FCC fee. Volunteer examiners may charge a small fee. Novice and Advanced licenses are no longer issued.

HF privileges by class (MHz)

Band	Technician	General	Amateur Extra
160 m	None	1.800–2.000	1.800–2.000
80 / 75 m	3.525–3.600 CW	3.525–3.600 CW, data 3.800–4.000 voice	3.500–3.600 CW, data 3.600–4.000 voice
60 m	None	5.3515–5.3665, plus 5.332, 5.348, 5.373, 5.405	Same as General
40 m	7.025–7.125 CW	7.025–7.125 CW, data 7.175–7.300 voice	7.000–7.125 CW, data 7.125–7.300 voice
30 m	None	10.100–10.150 CW, data	Same as General
20 m	None	14.025–14.150 CW, data 14.225–14.350 voice	14.000–14.150 CW, data 14.150–14.350 voice
17 m	None	18.068–18.110 CW, data 18.110–18.168 voice	Same as General
15 m	21.025–21.200 CW	21.025–21.200 CW, data 21.275–21.450 voice	21.000–21.200 CW, data 21.200–21.450 voice
12 m	None	24.890–24.930 CW, data 24.930–24.990 voice	Same as General
10 m	28.000–28.300 CW, data 28.300–28.500 voice	28.000–28.300 CW, data 28.300–29.700 voice	Same as General

All three classes get 6 m and up in full. General and Extra also get 2200 m and 630 m. Power: 200 W PEP on Technician HF segments and on 30 m, 100 W ERP on the four 60 m channels, 9.15 W ERP on the 60 m segment, and 1,500 W PEP elsewhere unless a band has a lower limit.

FRS and GMRS channels

Channels	Frequencies (MHz)	FRS	GMRS
1–7	462.5625 to 462.7125, 25 kHz steps	2 W	5 W
8–14	467.5625 to 467.7125, 25 kHz steps	0.5 W	0.5 W, handheld only
15–22	462.5500 to 462.7250, 25 kHz steps	2 W	50 W
Repeater inputs	467.5500 to 467.7250, 25 kHz steps	Not allowed	50 W

Test yourself

Question	Answer
What is the wavelength of 146 MHz, and what band is it?	About 2 m (300 ÷ 146). The 2 m ham band, in VHF.
Which services can you use with no test and no application?	FRS, MURS and CB. All are licensed by rule.
What does a GMRS license need that FRS does not?	An FCC application and $35. It has no test.
Which ham class first gets 20 m?	General.
Where on HF can a Technician use voice?	Only on 10 m, 28.300–28.500 MHz.
What is the difference between 118–137 and 225–400 MHz?	Both are AM airbands. The first is civil and the second is military.
Where does a 70 cm repeater usually listen, relative to its output?	5 MHz away, in either direction.
Does a scanner need a license?	Not for receiving, but some communications and some state laws carry limits.